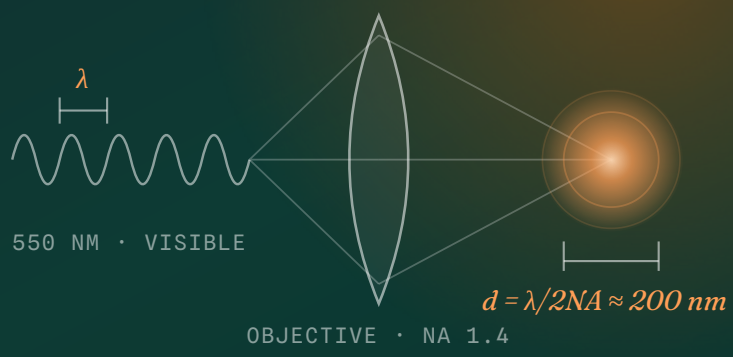
PRE-CONFERENCE WORKSHOP

Superresolution Microscopy

From Principles to Practice



Visible light cannot image a point smaller than this.

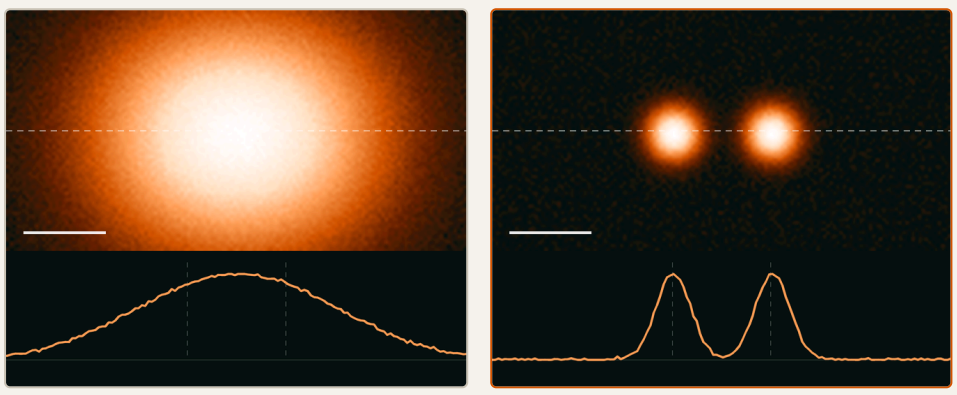
7–9 December 2026 • **IISER Pune & BRIC-NCCS Pune**

Hands-on with **Lattice SIM** and **STED** · Limited places, by application

Past the *diffraction* limit

Structures conventional imaging cannot separate

ONE SOURCE, OR TWO? point sources 120 nm apart

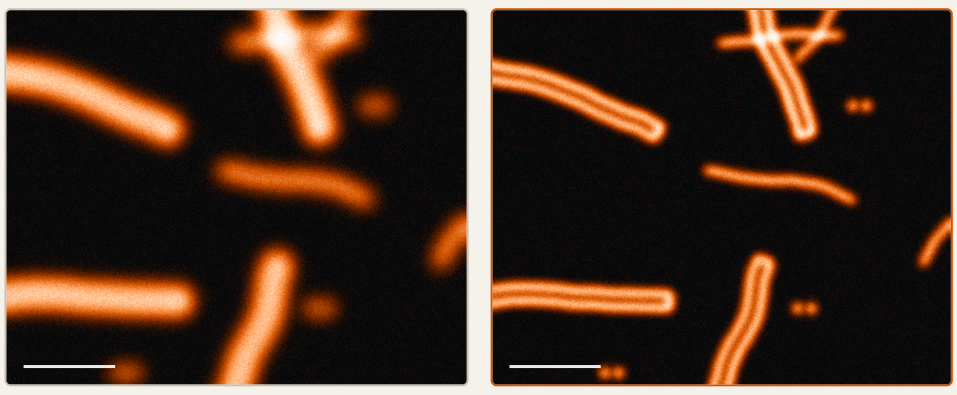


DIFFRACTION-LIMITED SUPERRESOLUTION

Below the diffraction limit, two sources are recorded as one. The dip between the peaks is **the resolution criterion**.

SCALE BAR 100 NM · SIMULATION

ONE FILAMENT, OR TWO? filaments 150 nm apart



DIFFRACTION-LIMITED SUPERRESOLUTION

Identical sample and label. What is recorded as a single filament is **two, 150 nm apart**.

SCALE BAR 1 µm · SIMULATION

Two approaches — **Lattice SIM** at IISER Pune, **STED** at BRIC-NCCS. Taught in batches of six by the scientists who operate them. **Every participant works on both.**

COURSE CONTENT

Lattice SIM

IISER PUNE

- Structured illumination from first principles
- **Multicolour, volumetric and live-cell acquisition**
- Reconstruction quality and its artefacts

STED

BRIC-NCCS PUNE

- Depletion beam alignment and quality
- **Tau-STED**: lifetime gating and live-cell imaging
- Fluorophore selection and labelling density

On both systems

SHARED SESSIONS

- Sample preparation and imaging controls
- **Quantitative image analysis**
- Calibration with fluorescent beads; reporting acquisition parameters

DATES
7–9 Dec 2026

VENUES
IISER Pune
BRIC-NCCS Pune

PARTICIPANTS
12
two batches of six

APPLICATIONS CLOSE
31 Oct 2026°

FEE
₹5,000
students & post-docs
₹10,000
faculty & professionals

INCLUDED
All meals and two nights' shared accommodation

ELIGIBILITY

Prior hands-on microscopy and image analysis is mandatory. Applicants from research laboratories and imaging facilities are especially encouraged. Full-time participation on all three days is required.

APPLY HERE

aicbc2026.iiserpune.ac.in/#workshop

INSTRUMENT PARTNERS

aicbc2026.workshop@gmail.com

Organised under the 49th All India Cell Biology Conference

° provisional — confirm on the website before applying